How do hashtags and emojis affect irony detection?

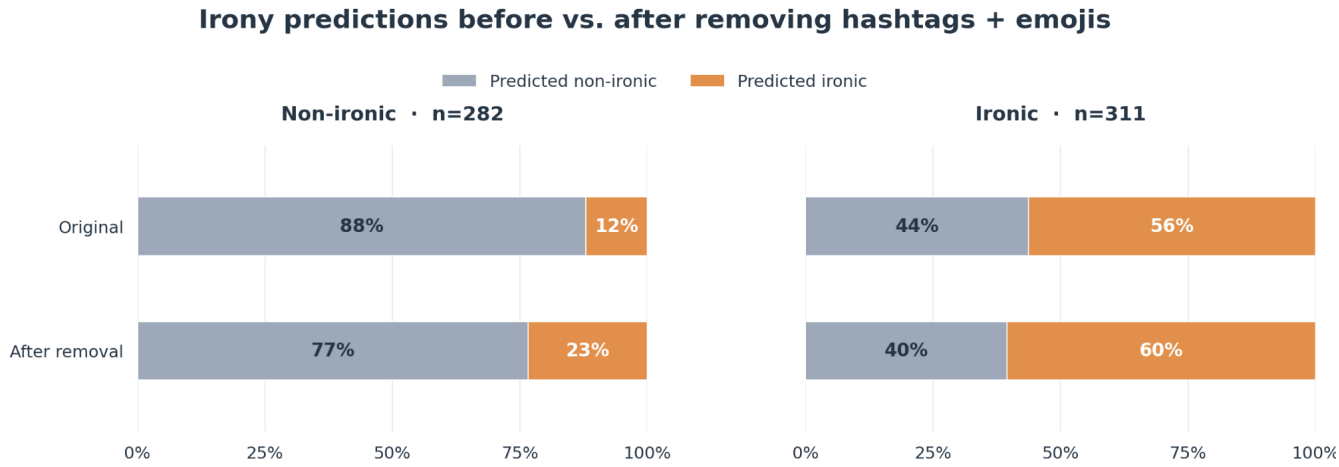
Phase I report · Three visualizations, results interpretation, and qualitative examples

Research question. How does Twitter-RoBERTa's irony detection behavior differ between ironic and non-ironic tweets when hashtags and emojis are removed?

Hypothesis. Removing hashtags and emojis changes the model's predicted irony labels and probabilities, potentially to different degrees for ironic and non-ironic tweets.

Experiment. Model: cardiffnlp/twitter-roberta-base-irony; dataset: TweetEval **irony/test** (784 tweets: 311 ironic, 473 non-ironic). We compared each tweet before and after removing hashtags, emojis, both, and only the # symbol. Comparisons use the **same tweets before and after** each modification.

FIGURE 1 · Predicted irony distribution before vs. after removal



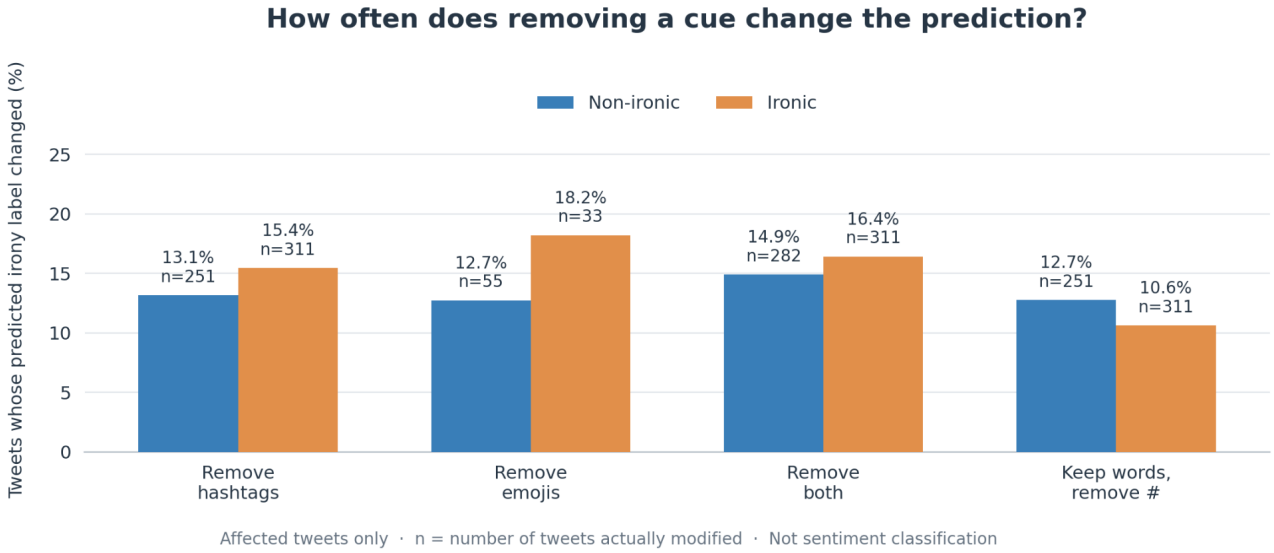
Only tweets containing a hashtag and/or emoji are included; before and after use the same tweets.

Reading the chart. Among affected ironic tweets, predictions of irony rise from **56% to 60%** after removing hashtags and emojis. Among affected non-ironic tweets, they rise from **12% to 23%**. Removing cues therefore increases the predicted irony share in both groups.

Each bar pair uses only tweets changed by combined removal (non-ironic n=282; ironic n=311). The chart shows **prediction distributions**; a higher irony rate is not always a better prediction.

How often does removing a cue change the predicted label?

FIGURE 2 · Label-flip rates across four controlled modifications



After hashtag removal, the model changes its prediction for **15.4% of ironic tweets** and **13.1% of non-ironic tweets** among affected inputs. The difference is **+2.3 percentage points** (bootstrap 95% CI **−3.4 to +8.0** points), so it does **not** establish a clear group difference. Removing both cues yields 16.4% versus 14.9%.

For emoji removal, the rates are **18.2% ironic** versus **12.7% non-ironic**; however, only 33 and 55 tweets are affected, and the interval for the difference includes zero. Emoji-related conclusions are therefore uncertain.

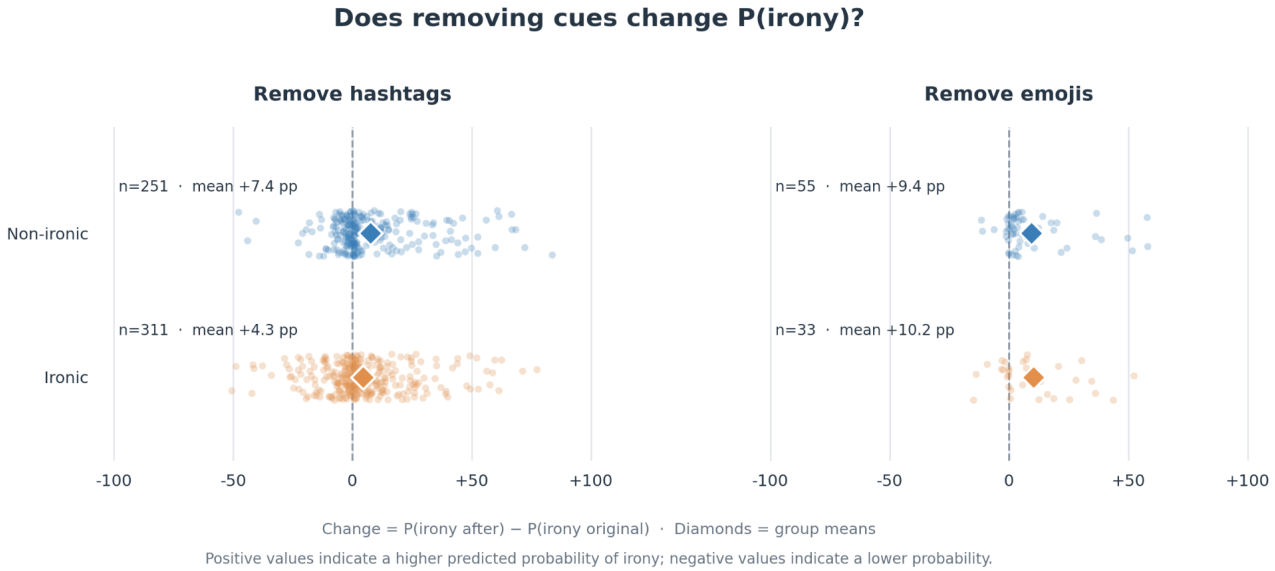
Critical follow-up: the groups differ in hashtag content. All ironic tweets (311/311) contain at least one of **#not, #irony, #sarcasm**, while only 113/473 non-ironic tweets contain one of these cues.

Affected subgroup	n	Hashtag-removal label flips
Ironic tweets (all include a cue)	311	15.4%
Non-ironic with #not / #irony / #sarcasm	113	12.4%
Non-ironic with other hashtags only	138	13.8%

When examined by hashtag subgroup, label-change rates are **15.4%, 12.4%, and 13.8%**, respectively. The main groups have different kinds of text, so we should not treat their comparison as a controlled effect of irony itself.

A prediction can change without changing its class

FIGURE 3 · Change in the probability assigned to irony



The horizontal axis is **P(irony after) − P(irony original)**, in percentage points. Each dot represents an affected tweet; the diamonds show group means. Values right of zero indicate a higher predicted probability of irony after removal.

Modification	Ironic mean ΔP(irony)	Non-ironic mean ΔP(irony)
Remove hashtags	+4.3 pp	+7.4 pp
Remove emojis	+10.2 pp	+9.4 pp

Interpretation. On average, hashtag removal increases the model's estimated probability of irony in both groups, more strongly for non-ironic tweets. Individual points may shift substantially even when the final class stays unchanged.

Caution. Removing whole hashtags also deletes their words (such as **#not**) and can change the meaning of the tweet. The control that removes only the **#** marker changes labels for 10.6% of affected ironic and 12.7% of affected non-ironic tweets.

Overall findings and limitations

Partial support for the hypothesis. Removing hashtags or emojis can alter binary irony predictions and their probabilities. The small group differences in flip rates, the unequal hashtag content, and the limited emoji samples prevent a strong claim that the effect is driven by irony itself. The ground-truth irony labels allow accuracy checks, but these visualizations focus on **black-box behavior**, not just benchmark performance.

Data source: uploaded **predictions.csv**, **statistics.csv** and **bootstrap.csv**. Figures are the updated Python-generated irony-model plots.

Three tweet-level examples

Cases from the correct irony detector, comparing the same tweet before and after full hashtag removal.

CASE 01 **Tweet 231 · Ironic** **Incorrect → correct**

ORIGINAL TWEET

i just enjoy throwing up #not #hatethis #sick #wanna #go #to #school #for #once #take #me #away

Predicted: **Non-ironic**

P(irony) **18.7%**

AFTER REMOVAL

i just enjoy throwing up

Predicted: **Ironic**

P(irony) **95.9%**

What this shows: Several hashtags, including #not, disappear. A large amount of text is removed, not just a marker.

CASE 02 **Tweet 307 · Ironic** **Incorrect → correct**

ORIGINAL TWEET

That's always the solution..... #not. Doesn't fix anything!

Predicted: **Non-ironic**

P(irony) **22.9%**

AFTER REMOVAL

That's always the solution..... . Doesn't fix anything!

Predicted: **Ironic**

P(irony) **94.0%**

What this shows: Removing #not raises the predicted irony probability; the resulting label matches the ground truth.

CASE 03 **Tweet 622 · Non-ironic** **Correct → incorrect**

ORIGINAL TWEET

#forgive #others #not #because they #deserve #forgiveness #you #peace love this from @user #words #wisdom ...

Predicted: **Non-ironic**

P(irony) **14.6%**

AFTER REMOVAL

they love this from @user ...

Predicted: **Ironic**

P(irony) **98.3%**

What this shows: Removing a long sequence of hashtags changes the meaning and produces a false irony prediction.